Inspector's Brief: Demo District (Surveillance Zone A)

Date: October 2026 | Surveillance: Sentinel-2 RGB (10 m/pixel) | Model: ChimneyCensus-OBB v1.0 (Rotated IoU mAP: 0.834) | Officer: Inspector Farida Khan

KEY SURVEILLANCE FINDINGS:
- 65 unique active kilns detected across 120 tiles (~154 sq km), deduplicated across 30 px tile overlap boundaries.
- 18 (15–21) baseline 2023–24 FCBKs appear converted to cleaner Zigzag designs (mandate-compliant).
- 27 persistent illegal FCBKs remain in operation post the 2022 ban; 8 circular kilns (CFCBK) require heritage review.
- 51 kilns are clustered within 1 km across 18 local groups, allowing Farida to inspect multiple sites per vehicle patrol.

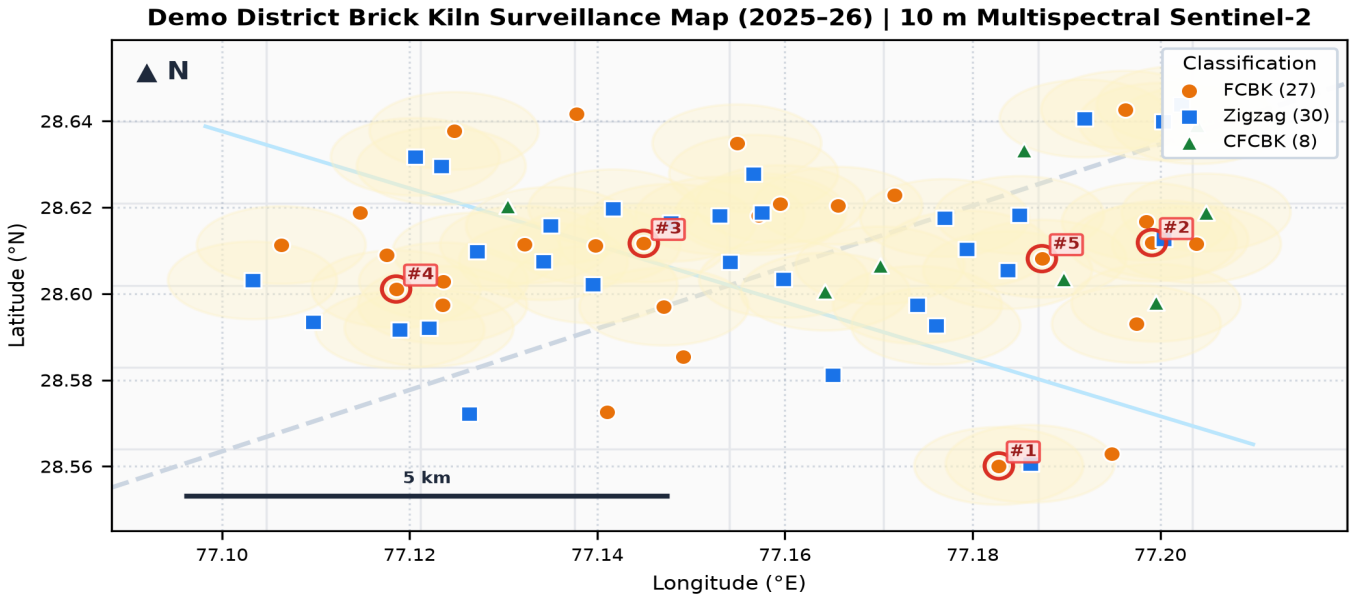


Table A: Kiln Counts by Design Type & Honest Uncertainty Bounds

Type	Detected	Est. True Count (95% CI)	Precision	Recall	Note
FCBK	27	18 – 41	0.89	0.85	Banned 2022 design
Zigzag	30	21 – 45	0.65	0.62	Mandate compliant
CFCBK	8	3 – 14	0.90	0.95	Circular kiln (~3%)
Total	**65**	**42 – 100**	**-**	**-**	**All types combined**

Method: Estimated true count adjusts raw detections for held-out validation precision and recall, using 1,000 bootstrap resamples for the 95% interval.

Table B: Conversion Status of Historical (2023–24) FCBK Kilns

Matched 2023–24 FCBKs	Now Detected as Zigzag (Low–High)	Still FCBK	Not Found / Unclear
45 (35 matched)	18 (15–21)	15 (Violation)	12

Conversion represents a 2023–24 registered FCBK site matched within ~30 m that now presents rectangular Zigzag geometry. Range reflects matching uncertainty.

Field Inspection Dispatch Plan & Top Priority Targets

Table C: Priority Proximity Clusters (Within 1 km of Neighbor) — Total: 51 kilns in 18 clusters. Full registry in brief/outputs/within_1km.csv.

Cluster	Kilns	Types	Centre lat/lon	Max gap in m
Cluster 1	5	3 FCBK, 1 Zigzag, 1 CFCBK	28.61431, 77.20135	958 m
Cluster 2	5	3 Zigzag, 1 CFCBK, 1 FCBK	28.64119, 77.19892	1,187 m
Cluster 3	4	2 FCBK, 2 Zigzag	28.61151, 77.13852	1,140 m
Cluster 4	4	3 FCBK, 1 Zigzag	28.61951, 77.15996	866 m
Cluster 5	3	3 FCBK	28.60046, 77.12191	634 m

Table D: Top 20 Kilns Farida Should Visit First (Actionable Dispatch Order)

#	Kiln ID	Type	Lat (°N)	Lon (°E)	Conf	Why visit
1	KILN-DEMO-025	FCBK	28.56011	77.18279	0.96	Still FCBK post-2022 ban; high conf (0.96); near 1 kiln
2	KILN-DEMO-053	FCBK	28.61186	77.19911	0.95	Still FCBK post-2022 ban; high conf (0.95); near 4 kilns
3	KILN-DEMO-048	FCBK	28.61167	77.14496	0.90	Still FCBK post-2022 ban; high conf (0.90); near 3 kilns
4	KILN-DEMO-056	FCBK	28.60109	77.11857	0.91	Still FCBK post-2022 ban; high conf (0.91); near 3 kilns
5	KILN-DEMO-052	FCBK	28.60813	77.18738	0.87	Still FCBK post-2022 ban; high conf (0.87); near 3 kilns
6	KILN-DEMO-030	FCBK	28.61137	77.13226	0.90	Still FCBK post-2022 ban; high conf (0.90); near 4 kilns
7	KILN-DEMO-032	FCBK	28.62078	77.15950	0.93	Still FCBK post-2022 ban; high conf (0.93); near 5 kilns
8	KILN-DEMO-023	FCBK	28.61803	77.15719	0.85	Still FCBK post-2022 ban; high conf (0.85); near 5 kilns
9	KILN-DEMO-028	FCBK	28.63773	77.12474	0.87	Still FCBK post-2022 ban; high conf (0.87); near 2 kilns
10	KILN-DEMO-047	FCBK	28.61114	77.13983	0.85	Still FCBK post-2022 ban (conf 0.85); near 6 kilns
11	KILN-DEMO-042	FCBK	28.61674	77.19852	0.84	Still FCBK post-2022 ban (conf 0.84); near 4 kilns
12	KILN-DEMO-057	FCBK	28.60287	77.12362	0.81	Still FCBK post-2022 ban (conf 0.81); near 4 kilns
13	KILN-DEMO-027	FCBK	28.59299	77.19744	0.81	Still FCBK post-2022 ban (conf 0.81); near 1 kiln
14	KILN-DEMO-044	FCBK	28.61129	77.10634	0.56	Still FCBK post-2022 ban (conf 0.56); near 1 kiln
15	KILN-DEMO-024	FCBK	28.59741	77.12353	0.71	Still FCBK post-2022 ban (conf 0.71); near 4 kilns
16	KILN-DEMO-039	FCBK	28.62041	77.16564	0.79	Still FCBK post-2022 ban (conf 0.79); near 4 kilns
17	KILN-DEMO-021	FCBK	28.64266	77.19630	0.80	Still FCBK post-2022 ban (conf 0.80); near 4 kilns
18	KILN-DEMO-059	FCBK	28.59700	77.14710	0.71	Still FCBK post-2022 ban (conf 0.71); near 1 kiln
19	KILN-DEMO-020	FCBK	28.63486	77.15489	0.72	Still FCBK post-2022 ban (conf 0.72); near 1 kiln
20	KILN-DEMO-054	FCBK	28.61150	77.20386	0.38	Unverified FCBK; low conf (0.38) check; near 4 kilns

Ranking score = Violation type (+40 FCBK, +35 CFCBK, +10 Zigzag) + Proximity (+20 within 1 km) + High confidence (+25 >= 0.85) or Verification urgency (+15 < 0.40).

HOW SURE ARE WE? (LIMITATIONS & FIELD GUIDANCE):

• 10 m Resolution: Sentinel-2 pixels span 10×10 m. Subtle internal draft retrofits inside oval walls may look like FCBKs; field checks resolve this.

• 2025–26 Domain Shift: Seasonal soil and atmospheric variations may cause minor classification drift compared to training imagery.

• No Ground Truth for Demo District: Estimates rely on held-out regional validation evidence; borderline cases (< 0.50 conf) require ground confirmation.